Blue Bloods in the NBA

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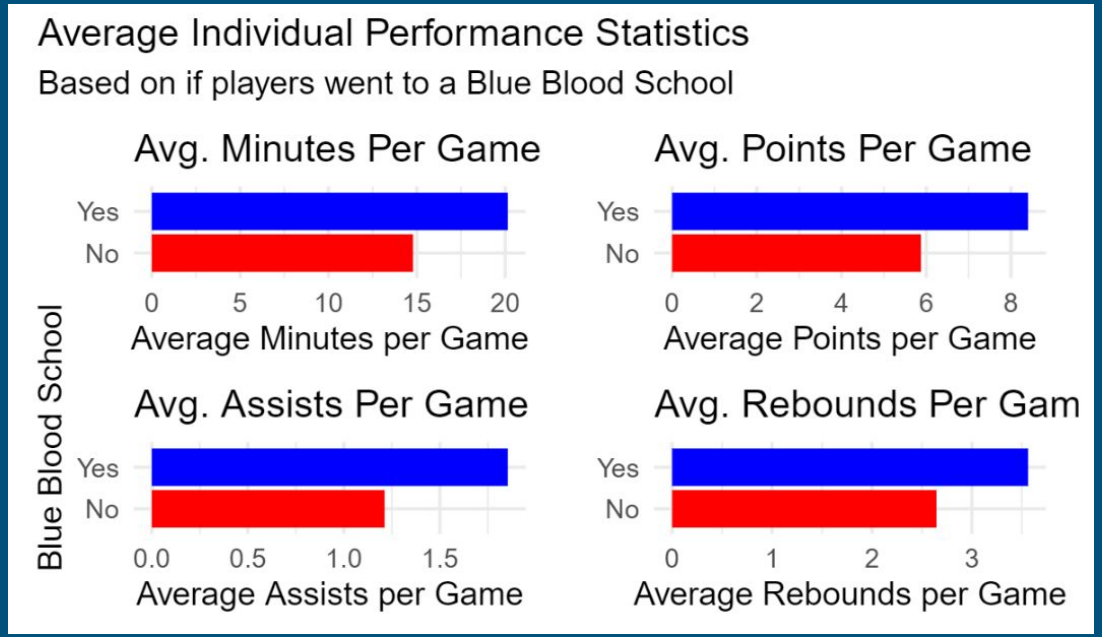
Topic and Motivation

The Blue Blood schools (Duke, Kentucky, Kansas, UNC, and UCLA) are some of the most successful college basketball programs. They are also the five most drafted from universities. We wanted to investigate if these schools live up to their reputation for producing elite NBA talent, and if players drafted from Blue Blood schools have better professional careers than those drafted alongside them.

Introduce the Data

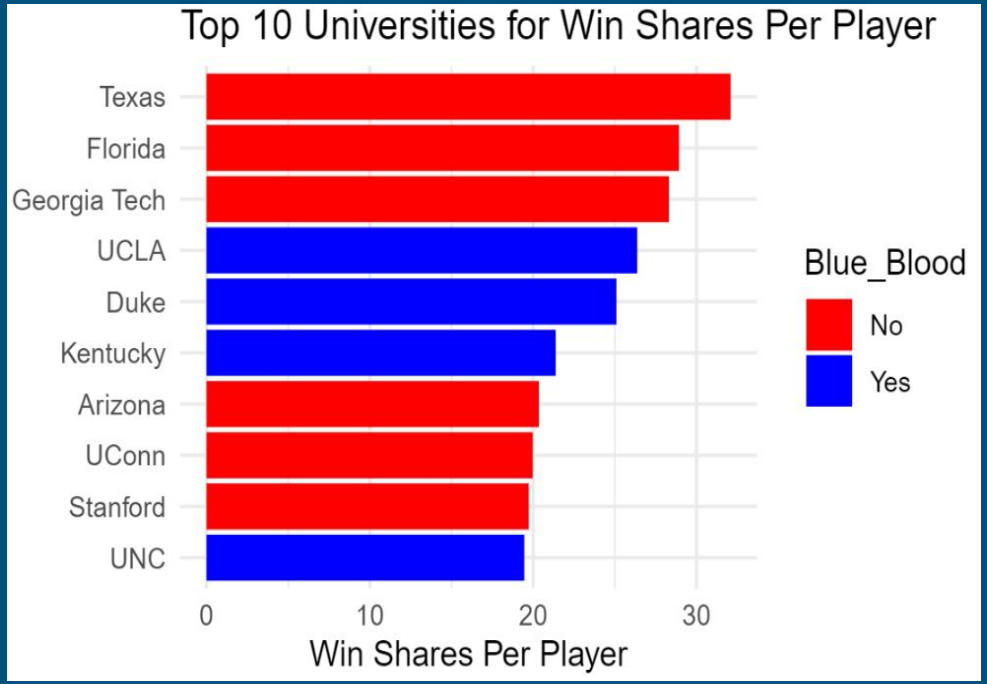
- The data set includes all players that have been drafted into the NBA from 2000-2016.
- Collected by Basketball Reference year by year as new players were drafted and as they have played.
- All these statistics are tracked by the NBA.
- Basketball Reference organizes their draft data by year so, to obtain a single data frame that included all drafts between 2000-2016, we downloaded the data for each year and combined it into a single data frame.
- Undrafted players are not included.

Box Score Analysis



We began our analysis by comparing basic box score production of Blue Blood players and other draftees. Blue Blood players are one the court more often, and they hold a significant edge in these three other major counting stats. However, looking purely at per game volume does not account for defensive impact, efficiency, or longevity, and this is why we conducted the majority of our analysis using advanced analytics.

Comparing Advanced Stats



We examined more advanced stats to get an all-encompassing picture of player success. We looked at the top universities in VORP, BPM, and Win Shares per drafted player. Blue Blood schools were near the top for each stat. We believed Win Shares to be the most indicative of career success, so we further explored that stat. As seen in the graphic to the left, four of the five Blue Blood schools are in the top 10 universities for win shares per player. The mean career win shares for a Blue Blood player was 21.51, and for all other players drafted it is 14.84.

Regression and Hypothesis Test

We created a linear model that predicted career win shares based on draft year and Blue Blood status. This model found that draftees from Blue Blood schools were expected to have 7.8 more career win shares. To prove that this difference is not by chance, we decided to conduct a simulation based two-proportion hypothesis test based on mean win shares.

$$H_0 : \mu_{BB} = \mu_N \text{ vs. } H_a : \mu_{BB} > \mu_N$$

Based on these hypotheses and our simulation, we found that the probability of seeing a difference in mean win shares of Blue Blood players and other draftees as large as the observed difference is 0.0045. This extremely small value allows us to reject the null hypothesis and make our final conclusion that players from Blue Blood schools have more successful NBA careers.

Conclusion

We have found significant evidence to say that Blue Blood Schools indeed produce stronger talent than average. Comparing both box score statistics and advanced statistics, we find that Blue Blood players have higher than average win shares per player, average points per game, average assists per game, average minutes per game, and average rebounds per game. This indicates that they are performing well individually. Using the results from the Hypothesis test, we find that players drafted from Blue Blood Schools have higher win shares in the NBA, indicating that Blue Blood players win more in the NBA and contribute more to team wins. Using these findings, Blue Blood players are individually better and have better careers than average players, which indicates Blue Blood Schools produce stronger talent.